Calculus I Assignment 4

1. If $f(x)$ is continuous on $[a, b]$ and $f(x) \neq 0$ for all $x \in [a, b]$, then $f(x)$ must be either strictly positive or strictly negative on $[a, b]$.

2. If a rock is thrown upward on the planet Mars with a velocity of 10 m/s, its height (in meters) after t seconds is given by $H = 10t - 1.86t^2$.

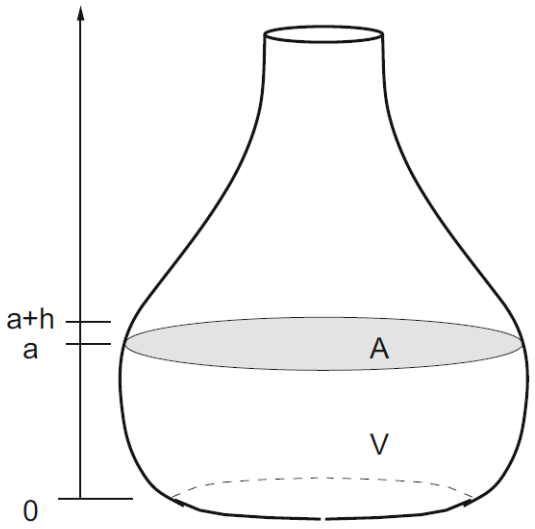
   (a) Find the velocity of the rock after one second.

   (b) Find the velocity of the rock when $t = a$.

   (c) When will the rock hit the surface?

   (d) With what velocity will the rock hit the surface?

3. The volume of water in a bottle is a function of the depth of the water. Let $V(a)$ be the volume up to depth a. Let $A(a)$ be the cross-sectional area of the bottle at height a.



   (a) What does $V(a + h) - V(a)$ represent? $(h > 0)$.

   (b) Suppose that a and $a + h$ are located as in the above figure. Put the following quantities in order:

$$A(a + h)h, \quad A(a)h, \quad V(a + h) - V(a)$$

   Write your answer using an inequality.

   (c) Using your inequality from part (b), explain why the quotient $\frac{V(a+h)-V(a)}{h}$ tends to $A(a)$ as h tends to 0.

4. For each function f given below, find $f'(a)$ by forming the difference quotient $\frac{f(a+h)-f(a)}{h}$, and taking the limit as h tends to 0. Then find the line tangent to the graph of f at $x = a$.

(a) $f(x) = \sqrt{x}, a = 4$.

(b) $f(x) = mx^2 + kx, a = 2$.

(c) $f(x) = x^3, a = -1$.

5. Show that a function that is not continuous at a cannot be differentiable at a.

6. The definition of differentiation implies that $f(x)$ is differentiable at x, if and only if

$$\lim_{\Delta x \to 0+} \frac{f(x + \Delta x) - f(x)}{\Delta x}$$

and

$$\lim_{\Delta x \to 0-} \frac{f(x + \Delta x) - f(x)}{\Delta x}$$

both exist and equal. These limits are called the **right-hand derivative** and **left-hand derivative** of f at x, and denoted by $f'_+(x)$ and $f'_-(x)$, respectively. In general, they may be different.

(a) Find the right-hand derivative of the function $y = y(x)$ at $x = 0$, where

$$y = \begin{cases} xe^{-1/x}, & x > 0, \\ 0, & x = 0. \end{cases}$$

(b) Find the left-hand and right-hand derivatives of $y = y(x)$ at $x = 0$, where

$$y = \begin{cases} x^2 \sin \frac{1}{x}, & x > 0, \\ \cos x - x - 1, & x \leq 0. \end{cases}$$

Determine whether this function is differentiable at the origin.

7. At which points is the function $y = \sqrt[3]{x^2}$ not differentiable? Why? Also, sketch the graph of this function.

8. Let $g(x) = x^2 \sin\left(\frac{\pi}{x}\right), g(0) = 0$.

(a) Sketch the graph of g on $-1 \leq x \leq 1$.

(b) Show that g is continuous at 0.

(c) Show that g is differentiable at 0 and find $g'(0)$.

9. Let $y = f(x)$ be an even function, and suppose that it is differentiable at $x = 0$. Prove that $f'(0) = 0$, and explain the geometric meaning.

Bonus Questions

1. Let
$$f(x) = |x - a|\varphi(x),$$
where φ is continuous with $\varphi(a) \neq 0$. Prove that f is not differentiable at $x = a$. What are $f'_-(a)$ and $f'_+(a)$?

2. Prove that
$$f(x) = \begin{cases} x^2, & x \text{ is rational,} \\ 0, & x \text{ is irrational.} \end{cases}$$
is differentiable only at $x = 0$.